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Observer-based sliding mode admittance control for ultrasound robot force-tracking in complex interaction environments

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ABSTRACT

Robotic ultrasound systems can improve imaging consistency and reduce operator workload by maintaining stable probe–tissue contact forces. However, variations in tissue stiffness and surface position often degrade force-tracking performance. This paper presents a disturbance observer–based sliding mode admittance control (DOSMAC) framework to address these challenges. Stiffness and position variations are uniformly modeled as lumped external disturbances, estimated online by a disturbance observer and compensated through sliding mode control. The resulting signal is introduced as a virtual input to the admittance model, enhancing robustness without requiring online stiffness identification or extensive parameter tuning. The stability of the closed-loop system is established through Lyapunov analysis. Simulations and experiments with stiffness ranging from 500 to 3500 N/m verify the effectiveness of the proposed approach. In experiments with simultaneous stiffness and position variations, DOSMAC reduces overshoot by 29.1% and 55.4%, peak force error by 4.36 N and 0.46 N, and root mean square error by 1.79 N and 0.4 N, respectively, compared with conventional admittance control and adaptive variable admittance control. These results demonstrate that the proposed method enables stable and reliable force tracking under complex time-varying conditions, supporting the clinical translation forceof robotic ultrasound.

1. Introduction

Robotic ultrasound inspection has attracted increasing attention in recent years due to its potential to standardize imaging quality and reduce operator workload in clinical examinations (Li et al., 2021; Wu et al., 2025). A key aspect of robotic assistance lies in its ability to maintain a stable and controlled contact force between the ultrasound probe and human tissue, which is crucial for ensuring reliable acoustic coupling and consistent image resolution (Ihnatsenka & Boezaart, 2010). Unlike manual operation, where variations in applied force can degrade image quality and depend heavily on the operator's skill and endurance, robotic systems offer the promise of automation and repeatability (Huang et al., 2018), which are critical for long-duration or high-precision diagnostic tasks.

Active compliance strategies, such as impedance control (Hogan, 1984) and admittance control, also referred to as position-based impedance control (Seraji, 1994), have shown effectiveness in contact force tracking in robotic ultrasound systems (Bao et al., 2023; Mathiassen et al., 2016). However, maintaining a stable contact force in practice remains a nontrivial problem, particularly in complex environments such as human tissues with varying and uncertain mechanical properties (Fu et al., 2025). Human tissue stiffness can vary significantly across anatomical regions and change dynamically during scanning due to physiological movements such as breathing (Duan et al., 2025) or involuntary muscle activation (Cheng & Tavakoli, 2018). These variations introduce substantial uncertainty into the interaction dynamics, posing challenges for traditional control approaches. Effective force control under these conditions calls for robust strategies that can accommodate rapid stiffness fluctuations without compromising performance or safety.

To tackle these challenges, numerous methods have been explored. Variable impedance control has been shown to enhance adaptability in uncertain environments, with adaptive (Duan et al., 2018), optimization-based (Jin et al., 2022; Wang et al., 2022), and

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learning-based approaches (Hamedani et al., 2021; Yoon et al., 2024) proposed to adjust impedance parameters online. While effective, these strategies often require extensive tuning, computationally intensive optimization, or large datasets, which may reduce practicality in highly dynamic clinical conditions. Energy tank–based frameworks (Ferraguti et al., 2015) have been introduced to preserve passivity during parameter adaptation and have been applied to robotic ultrasound (Beber et al., 2024; Fu et al., 2025), but typically impose additional architectural and computational overhead.

Robust control approaches based on sliding-mode admittance control have also been investigated (Solanes et al., 2018; Torabi et al., 2018). For instance, sliding-mode regulation embedded in admittance control enables fast response and robustness to disturbances (Zhou et al., 2025). However, accurate knowledge of the environment stiffness is usually required, limiting their suitability for tissue interactions with rapidly time-varying mechanical properties.

Disturbance-observer based control methods, including those integrated with sliding-mode control (DOB–SMC), offer an alternative means of handling model uncertainties (Dao & Ahn, 2022; Wu et al., 2019; Zhang et al., 2025a,b). These methods estimate unknown or time-varying dynamics as lumped disturbances and compensate them through sliding-mode regulation.

For instance, Huo et al. (2025) proposed an observer-based adaptive robust force controller in which the low-level motion-tracking error is treated as an input disturbance of the high-level force loop. Their method integrates an extended state observer with adaptive parameter estimation and achieves improved force-tracking performance in uncertain conditions. Though, some environment uncertainties are implicitly compensated, they focuses mainly on compensating motion-tracking errors, without explicitly addressing strongly time-varying environment stiffness or shifting equilibrium positions.

Nonetheless, existing DOB–SMC approaches exhibit three main limitations with respect to robotic ultrasound:

(i) Existing DOB–SMC methods mainly estimate robot-side dynamic disturbances (e.g., inertia mismatch, friction) and do not explicitly address dual time-varying soft-tissue behavior, where both stiffness and equilibrium position change concurrently.
(ii) Most DOB–SMC schemes are developed for rigid or weakly time-varying environments (industrial contact, polishing, rehabilitation) and have not been designed or validated for ultrasound imaging, where tissue deformation and surface motion dominate.
(iii) Conventional DOB–SMC controllers are typically implemented in joint-space or impedance frameworks, and are not integrated into admittance control, limiting applicability to position-controlled robots commonly used in clinical ultrasound.

To overcome these limitations, a disturbance-observer based sliding-mode admittance control (DOSMAC) framework is developed for robotic ultrasound applications. The proposed approach reconstructs lumped environmental uncertainties—including stiffness variations, equilibrium position drift, and modeling discrepancies—through a disturbance observer, while a sliding-mode regulation ensures robust force tracking under rapidly changing interaction conditions. This design achieves a favorable balance between robustness, computational efficiency, and practical implementability, enabling real-time deployment on clinical robotic platforms.

The main contributions of this paper are as follows:

1. A robust and computationally efficient admittance control framework (DOSMAC) is developed, in which environmental stiffness variations and equilibrium-position deviations are reformulated as a lumped disturbance, eliminating the need for real-time stiffness estimation or extensive parameter tuning.
2. A rigorous Lyapunov-based stability analysis is established, demonstrating boundedness of all closed-loop signals and asymptotic convergence of the force-tracking error in the presence of unknown, time-varying stiffness and contact-position uncertainties.

Comprehensive simulations and experiments further demonstrate the effectiveness and robustness of the proposed method across a wide range of stiffness and surface-motion conditions

The remainder of this paper is organized as follows. Section 2 formally defines the force control problem. Section 3 describes the proposed disturbance observer–based sliding mode admittance control (DOSMAC) strategy, along with its implementation details, followed by a Lyapunov stability analysis. Simulation results are provided in Section 4, followed by experimental validations in Section 5. Section 6 presents results and discussions of this paper. Conclusions with future research directions are presented in Section 7.

Symbol	Definition
x, x_d, x_c	Actual, reference and commanded position in the controlled direction
f_{ext}, f_d	Measured and desired contact force
$\boldsymbol{M}$, $\boldsymbol{B}$, $\boldsymbol{K}$	Virtual mass, damping, and stiffness matrices
m, b, k	One-dimensional virtual mass, damping, stiffness
k_e	Environment stiffness
k_{e0}	Nominal stiffness used for gain normalization
x_e	Environment equilibrium position
$\hat{x}_e$	Estimated environment position (first-contact estimate)
δx_e	Environment position error
$e = x - x_e$	Penetration depth
$\hat{e}$	Estimated penetration depth, $\hat{e} = x - \hat{x}_e$
y	Force-tracking error, $y = F_{ext} - F_d$
s	Sliding variable, $s = Cy + \dot{y}$
$\hat{s}$	Estimated sliding variable in DOB
$\tilde{s}$	Sliding-variable estimation error
C	Sliding gain (convergence rate)
k_s	Sliding-mode gain for reaching condition
η	Boundary-layer gain for chattering mitigation
ϕ	Boundary-layer thickness for the saturation function
$\mathrm{sat}(\cdot)$	Saturation function
u	Virtual control input (generated by DOSMAC)
G_0	Nominal virtual input gain
ΔG	Input-gain mismatch, $\Delta G = -\frac{k_e - k_{e0}}{m}$
$d(t)$	Lumped disturbance (uncertainties from stiffness/position)
$\hat{d}(t)$	Disturbance estimated by DOB
$\tilde{d}(t)$	Disturbance estimation error, $\tilde{d}(t) = d(t) - \hat{d}(t)$
L_1, L_2	Observer gains of the DOB
h_{nom}	Nominal measurable term in sliding dynamics
V, V_1, V_2	Lyapunov functions for stability analysis
T_s	Sampling period (controller update interval)
τ	Robot control cycle (UR5e: 2 ms)
t_c, t_s	Contact time, stabilization time

2. Problem statement

2.1. Robot-human body interaction model during robotic ultrasound

Fig. 1 illustrates the interaction between a robotic ultrasound probe and human body during a typical ultrasound examination. The probe is rigidly mounted on the robot's end-effector flange and establishes contact with the body surface as the robot follows the scanning trajectory. The interaction force generated at the probe–tissue interface is regulated by a force control algorithm, most commonly an admittance controller.

To analyze this interaction, Fig. 1(b) presents a simplified one-dimensional model along the z-axis. In this representation, the human

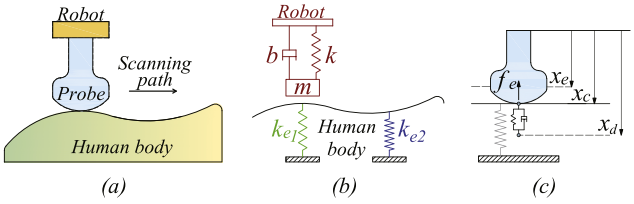


Fig. 1. (a) Illustration of the robot-environment interaction during robotic ultrasound (b) the interaction model of admittance control and human body environment (c) contact description of admittance model.

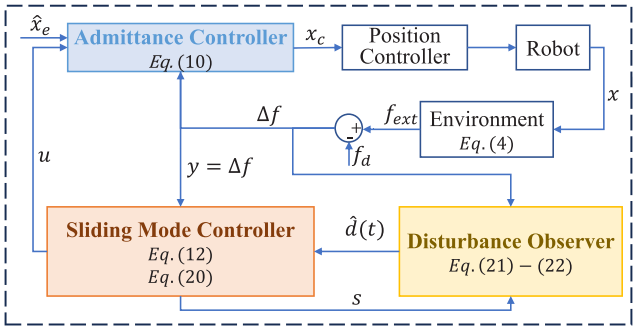


Fig. 2. Control scheme of the proposed DOSMAC method. $\hat{x}_e$ is the position where contact first starts.

body is modeled as a linear spring characterized by a time-varying equilibrium position and stiffness. When the probe touches the body surface, as shown in Fig. 1(c), the controller measures the interaction force and receives the desired trajectory. Based on these inputs and the chosen impedance parameters, it generates a commanded trajectory, which is subsequently executed by the robot's internal position controller to achieve compliant behavior.

As depicted in Fig. 1(c), the admittance controller introduces a virtual second-order mass–spring–damper system between the reference trajectory x_d and the commanded trajectory x_c. By converting the measured contact force into a compliant motion command, admittance control enables force regulation even on robots that lack native force-control modes (Duan et al., 2018).

The classical admittance model describes how the robot motion responds to external forces through a virtual dynamic system. This formulation provides the foundation for converting force deviations into corrective position commands.

For a manipulator operating in an n-dimensional Cartesian space, the classical admittance law is expressed as:

$$M(\ddot{X}_c - \ddot{X}_d) + B(\dot{X}_c - \dot{X}_d) + K(X_c - X_d) = F_{ext} \tag{1}$$

where $M \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times n}$ and $K \in \mathbb{R}^{n\times n}$ are symmetric, positive-definite virtual mass, damping, and stiffness metrics, respectively. The vectors $X_d, X_c \in \mathbb{R}^n$ are the reference and command trajectories. The vector $F_{\text{ext}} \in \mathbb{R}^n$ is the external force measured by the force/torque sensor mounted at the tool flange.

Because each Cartesian axis can be treated independently for typical soft-tissue contact, the subsequent analysis considers a one-dimensional case without loss of generality.

For one-dimensional motion, the admittance model reduces to:

$$m(\ddot{x}_c - \ddot{x}_d) + b(\dot{x}_c - \dot{x}_d) + k(x_c - x_d) = \Delta f \tag{2}$$

where

$$\Delta f = f_{ext} - f_d \tag{3}$$

Here, m, b, and k are the one-dimensional virtual mass, damping and stiffness coefficients, respectively. And f_d denotes the desired contact force.

The environment is modeled as a linear spring:

$$f_{ext} = -k_e(x - x_e) \tag{4}$$

with k_e representing the environment stiffness and x_e its equilibrium position. Because industrial and collaborative robots typically exhibit high-precision position tracking, it is assumed that there is no error between the actual robot position x and the command position x_c, meaning $x \approx x_c$. In the admittance architecture, the commanded motion x_c is directly executed by the robot's internal position controller, so no differentiation of x_c is required. Any tracking error is absorbed into the lumped disturbance and compensated within the DOSMAC framework.

The relationship between the desired trajectory and the achieved contact force can be derived by rearranging the admittance dynamics to reveal the motion needed to generate the desired force.

From Eq. (4), achieving the desired force requires: $x_d = -f_d/k_e + x_e$. This force–motion mapping is valid under the standard assumptions of admittance control, including known contact direction, stable probe–tissue engagement, and low-frequency interaction dynamics. However, since environment stiffness k_e is generally unknown and may vary with time or tissue properties, this trajectory is not directly available. A common workaround is to approximate x_d using the estimated equilibrium position $\hat{x}_e$, obtained from force sensing when contact is first detected. This approximation reflects the practical limitation that the true equilibrium position is not directly measurable in real robotic ultrasound interactions.

As shown in Jung et al. (2004), setting the virtual stiffness to zero, i.e., $k = 0$, allows the admittance controller to track the desired force for constant k_e. The resulting model becomes:

$$m\ddot{e} + b\dot{e} = \Delta f \tag{5}$$

where $e = x - x_e$ denotes the penetration depth. Substituting Eqs. (3) and (4) into Eq. (5), yields:

$$m\ddot{e} + b\dot{e} + k_e e = -f_d \tag{6}$$

Under constant k_e, properly chosen m and b may ensure stability. However, for time-varying stiffness (common in soft-tissue contact), the performance of the zero-stiffness admittance deteriorates significantly.

When the environment equilibrium position is time-varying, one has $\dot{x}_e \neq 0$ and $\ddot{x}_e \neq 0$. Let the estimated position be:

$$\hat{x}_e = x_e - \delta x_e \tag{7}$$

where δx_e represents the deviation between the assumed and actual equilibrium position and captures uncertainties caused by tissue deformation, probe tilt, or minor surface motion during scanning. This term is not explicitly estimated. Instead, its effect is incorporated into the lumped disturbance and compensated through the disturbance observer. Then:

$$e = x - x_e = (x - \hat{x}_e) - \delta x_e = \hat{e} - \delta x_e \tag{8}$$

In practice, only $\hat{e}$ is available. Replacing e with $\hat{e}$ in (6) yields the implemented dynamics:

$$m\ddot{\hat{e}} + b\dot{\hat{e}} = m(\ddot{e} + \delta\ddot{x}_e) + b(\dot{e} + \delta\dot{x}_e) = f_{ext} - f_d \tag{9}$$

This expression shows that estimation errors δx_e introduce additional disturbance terms that degrade force-tracking performance, especially when both k_e and x_e vary with time. These disturbance-like terms directly affect force-tracking performance, motivating the use of robust compensation mechanisms in the following sections.

3. Proposed method

This section presents the proposed Disturbance-Observer based Sliding-Mode Admittance Control (DOSMAC) framework, followed by the controller formulation and stability analysis.

3.1. System description

Section 2 established that traditional admittance control exhibits degraded force-tracking performance in complex interaction environments—such as those encountered in robotic ultrasound—where both environmental stiffness and equilibrium position vary with time. To enhance force-tracking robustness, a virtual control input u is introduced into the admittance dynamics:

$$m\ddot{\hat{e}} + b\dot{\hat{e}} = \Delta f + u \tag{10}$$

The virtual input u may include any measurable information, such as position or force-error-related terms (e.g., $u = \Delta b\dot{\hat{e}}$ yields a variable-parameter admittance controller). This virtual input provides additional flexibility to modify the admittance response and serves as an interface for incorporating robustness through sliding-mode control and disturbance estimation.

In the proposed framework, u is generated by a sliding-mode controller (SMC), while an extended-state-observer-like disturbance observer (DOB) compensates unknown environmental dynamics. The overall structure is illustrated in Fig. 2.

3.2. Controller design

Sliding manifold

To regulate the force-tracking error with a prescribed convergence behavior, a sliding manifold is defined. Define the force-tracking error as:

$$y = f_{ext} - f_d \tag{11}$$

A first-order sliding manifold is defined as:

$$s = Cy + \dot{y} \tag{12}$$

where $C > 0$ shapes the desired convergence rate. On the sliding manifold $s = 0$, the force error satisfies $\dot{y} = -Cy$, ensuring asymptotic convergence. This structure imposes first-order convergence on the force error once the system reaches the sliding manifold.

Desired sliding dynamics

To enforce sliding-surface reaching, the following target dynamics are imposed:

$$\hat{s} = -k_s s - \eta sat(s) \tag{13}$$

where $k > 0$ defines the linear convergence speed, and $\eta > 0$ ensures robustness to disturbances.

Sliding variable dynamics

Differentiating (12) gives:

$$\dot{s} = C\dot{y} + \ddot{y} = C\dot{y} + (\ddot{f}_{ext} - \ddot{f}_d) \tag{14}$$

In robotic ultrasound applications, the desired force is typically constant, implying $\ddot{f}_d = 0$. Using $f_{ext} = -k_e e$, the expression becomes:

$$\dot{s} = C\dot{y} + (-\ddot{k}_e e - 2\dot{k}_e - k_e\ddot{e}) \tag{15}$$

From equation (10), the admittance model yields:

$$\ddot{e} = \frac{1}{m}(y + u - b\dot{e} - (m\delta\ddot{x}_e + b\delta\dot{x}_e)) \tag{16}$$

Substituting this into equation (15) gives:

$$\dot{s} = C\dot{y} + (-\ddot{k}_e e - 2\dot{k}_e\dot{e} - \frac{k_e}{m}(y + u - b\dot{e} - (m\delta\ddot{x}_e + b\delta\dot{x}_e)) \tag{17}$$

Because k_e and δx_e are unknown and time-varying, this form is not directly suitable for solving u.

Therefore, all unknown terms are grouped into a lumped disturbance d, leading to:

$$\dot{s} = G_0 u + h_{nom} + d(t) \tag{18}$$

where $G_0 = -\frac{k_{e0}}{m}$ is the nominal input gain, $h_{nom} = C\dot{y}$ is a measurable nominal term, d includes stiffness variation, position-estimation errors, and input-gain mismatch:

$$d(t) = -\ddot{k}_e e - 2\dot{k}_e e - \frac{k_e}{m}y + \frac{k_e}{m}b(\dot{e} + \delta\dot{x}_e) + m\delta\ddot{x}_e + \Delta G u \tag{19}$$

where $\Delta G = -\frac{k_e - k_e 0}{m}$ quantifies the virtual input gain mismatch arising from variations in the environment stiffness relative to the nominal value k_{e0}. This mismatch is treated as part of the lumped disturbance and compensated by the DOSMAC framework.

Grouping these unknown and time-varying quantities into a single lumped disturbance allows the controller to estimate and compensate their combined effect without requiring explicit modeling of each uncertainty source.

Sliding-Mode Control Law

Now, the virtual input u can be obtained by combining Eq. (13) and (18):

$$u = \frac{1}{G_0}(-k_s s - \eta sat(s) - \hat{d} - h_{nom}) \tag{20}$$

where $\hat{d}$ is the disturbance estimate generated by the DOB.

Disturbance Observer

An ESO (Extended State Observer)-like observer is used to estimate and the lumped disturbance:

$$\dot{\hat{s}} = G_0 u + h_{nom} + \hat{d} + L_1(s - \hat{s}) \tag{21}$$

$$\dot{\hat{d}} = L_2(s - \hat{s}) \tag{22}$$

where L_1 and L_2 are the observer gains. This observer adopts an extended-state-observer structure, which enables online reconstruction of the lumped disturbance and the sliding variable dynamics using only measurable signals.

3.3. Stability analysis

This subsection establishes the closed-loop stability of the proposed DOSMAC framework. The analysis first examines the disturbance observer and sliding-mode subsystems individually and then employs a composite Lyapunov function to demonstrate overall boundedness and asymptotic convergence of the force-tracking error.

Theorem 1 (Stability of the DOSMAC Closed-Loop System).

Consider the admittance-controlled robot interacting with a time-varying environment described by the dynamics in (10)–(22).

Assume that:

1. the lumped disturbance d and its derivative $\dot{d}$ are bounded,
2. the observer gains $L_1, L_2 > 0$, and L_2 is large enough that $\frac{\dot{\hat{d}}}{L_2}$ could be omitted,
3. the sliding-mode gain η satisfies $\eta \geq |\tilde{d}|$, where $\tilde{d} = d - \hat{d}$ is the disturbance estimation error.

Then the closed-loop signals are bounded, the sliding variable s converges to zero, and the force-tracking error $y = f_{ext} - f_d$ converges asymptotically to zero.

Thus, the proposed DOSMAC admittance controller guarantees stable force-tracking performance.

Proof.

1. **Disturbance Observer (DOB) Stability**

   The first step shows that the observer ensures bounded estimation errors under the given gain conditions.

   Define the Lyapunov function for the DOB subsystem as:

$$V_1 = \frac{1}{2L_2}\tilde{d}^2 + \frac{1}{2}\tilde{s}^2 \tag{23}$$

where $\tilde{d} = d - \hat{d}$ and $\tilde{s} = s - \hat{s}$ are the estimation errors of the lumped disturbance and sliding variable respectively.

Differentiating (23) gives:

$$\dot{V}_1 = \frac{1}{L_2}\dot{\tilde{d}}\tilde{d} + \dot{\tilde{s}}\tilde{s} \tag{24}$$

For the first term in V_1, one has:

$$\frac{1}{L_2} = \frac{1}{L_2}(\dot{d} - \dot{\hat{d}})\tilde{d} = (\frac{\dot{d}}{L_2} - \frac{\dot{\hat{d}}}{L_2})\tilde{d} \tag{25}$$

Substituting Assumption 2 yields:

$$\frac{1}{L_2}\dot{\hat{d}}\tilde{d} = -\frac{1}{L_2}\dot{\tilde{d}}\tilde{d} \tag{26}$$

Substituting Eq. (22) leads to:

$$\frac{1}{L_2}\dot{\hat{d}}\tilde{d} = -\frac{1}{L_2}\dot{\hat{d}}\tilde{d} = -\frac{1}{L_2}(L_2(s - \hat{s}))\tilde{d} = -\tilde{s}\tilde{d} \tag{27}$$

Substituting this result back into V_1, one obtains:

$$\dot{V}_1 = -\tilde{s}\tilde{d} + \dot{\tilde{s}}\tilde{s} \tag{28}$$

Taking the difference between Eqs. (18) and (21) one has: $\dot{\tilde{s}} = \tilde{d} - L_1\tilde{s}$. Substituting into the above expression yield:

$$\dot{V}_1 = -\tilde{s}\tilde{d} + (\tilde{d} - L_1\tilde{s})\tilde{s} = -L_1\tilde{s} \le 0 \tag{29}$$

Thus, the estimation errors $\tilde{s}$ and $\tilde{d}$ remain bounded, and the DOB subsystem is stable.

2. **Sliding-Mode Control Stability**

The second step demonstrates that the sliding variable converges to zero under the proposed control law.

Consider the Lyapunov function:

$$V_2 = \frac{1}{2}s^2 \tag{30}$$

Taking the derivative:

$$\dot{V}_2 = s\dot{s} = s(G_0u + h_{nom} + d) \tag{31}$$

Substituting the control law (20):

$$\dot{V}_2 = \tilde{d}s - \eta|s| - k_s s^2 \tag{32}$$

the condition $\eta \ge |\tilde{d}|$ in assumption 3 yields:

$$\dot{V}_2 \le -k_s s^2 \le 0 \tag{33}$$

indicating that the sliding variable s converges to zero.

3. **Convergence of Force Error**

The sliding variable is defined as:

$$s = cy + \dot{y} \tag{34}$$

once $s \to 0$, the reduced-order dynamics satisfy:

$$\dot{y} = -cy \tag{35}$$

which guarantees: $y(t) \to 0$ as $t \to \infty$. Thus, the force-tracking error converges asymptotically.

4. **Overall Stability**

Combining both subsystems using a composite Lyapunov function yields the closed-loop stability result.

The composite Lyapunov function:

$$V = V_1 + V_2 \tag{36}$$

satisfies:

$$\dot{V} = \dot{V}_1 + \dot{V}_2 \le 0 \tag{37}$$

□

Therefore, all closed-loop signals remain bounded, and the DOSMAC system achieves stable and robust force tracking despite unknown time-varying environmental stiffness and position uncertainties.

Although the formulation is presented in a single translational DOF, this does not limit the applicability of the proposed framework. In standard Cartesian-space admittance control, the virtual inertia, damping, and stiffness are typically chosen diagonal, resulting in decoupled dynamics along each Cartesian axis. Under this common assumption, the DOSMAC controller can be directly extended to multi-DOF settings by applying the same control law independently to each axis, with the same structure and stability properties.

In robotic ultrasound applications, the variations of $k_e(t)$ and $x_e(t)$ are typically low-frequency, and the virtual control input is smoothed by the boundary-layer saturation function. As a result, the lumped disturbance and its derivative remain bounded and do not invalidate the assumption $\dot{d}/L_2$ used in the high-gain disturbance observer analysis.

Algorithm 1 DOSMAC Algorithm.

Initialization:
Set controller and observer parameters $(m, b, C, k_s, \eta, \phi, L_1, L_2)$ and the sampling period T_s.
Initialize observer states $\hat{d}(t_0)$, $\hat{s}(t_0)$, and nominal gain G_0.
while the control loop is active **do**
  Read force sensor $f_{ext}(t)$ and compute force error $y(t) = f_{ext}(t) - f_d(t)$ and $\dot{y}(t)$.
  Compute sliding variable $s(t) = C\,y(t) + \dot{y}(t)$ according to Eq. (12).
  Compute nominal term $h_{nom}(t) = C\,\dot{y}(t)$.
  Compute virtual input $u(t)$ via Eq. (20) with $\hat{d}(t-1)$.
  Update commanded motion $x_c(t)$ by integrating the admittance dynamics in Eq. (10).
  Update the disturbance observer states $\hat{d}(t)$ and $\hat{s}(t)$ using Eqs. (21) and (22).
end while

4. Simulation validation

4.1. Simulation model and settings

To verify the effectiveness and advances of the proposed DOSMAC method, simulation experiments are conducted based on a one-dimensional robot-environment interaction. The environment interaction is modeled according to Eq. (4), and the robot position tracking error of the controller is omitted. The output of the admittance controller, x_c, is directly considered as the actual robot position x.

In simulation, the true environment position x_e is directly used as the nominal value $\hat{x}_e$ to isolate the evaluation of controller performance from position-estimation errors. The effect of any potential mismatch $\delta x_e = x_e - \hat{x}_e$ is therefore not simulated and is implicitly handled through the lumped disturbance term in the theoretical formulation.

To comprehensively evaluate the performance of the proposed method, three admittance control methods are compared through simulation.

AC: Traditional admittance controller with constant parameters, as described in Eq. (5).

AVAC: Adaptive variable admittance controller which adaptively adjusts damping parameters online according to force error information (Duan et al., 2018). The damping parameter is defined as:

$$b_{true} = b + \frac{1}{\dot{e}}b\Phi(t) \tag{38}$$

$$\Phi(t) = \Phi(t-1) - \frac{\sigma}{b}\Delta f(t-1) \tag{39}$$

DOSMAC: The proposed DOSMAC controller.

All three admittance controllers share the same admittance parameters: $m = 1, b = 200, k = 0$. For AVAC, the $\sigma = 0.1$ is chosen for balanced performance. For the proposed DOSMAC controller, the nom-

inal environment stiffness $k_0 = 1000\,\mathrm{N/m}$ is given, which is not fine-tuned. The rest of the parameters of the sliding mode controller are given as: $C = 25, k_s = 50, \eta = 10, \varphi = 0.1$. The DOB parameters $L_1 = 60$ and $L_2 = 1200$ are given for fast disturbance estimation. The control cycle $\tau = 0.002\,\mathrm{s}$ is set according to the real UR5e robot used in the experiment. The simulation lasts for 10 s for each test. All control strategies are implemented within MATLAB 2023b environment.

The following scenarios are considered to evaluate the adaptability of the controllers:

Scenario 1 (Varying Stiffness): Environment stiffness varies as $k_e = 3000 \times (1.5 + sin(0.2\pi t))N/m$.

Scenario 2 (Varying Position): Environment position changes as $x_e = 0.02 \times \sin(0.2\pi t)$ m.

Scenario 3 (Simultaneous Stiffness and Position Variation): Simultaneous variations in environment stiffness and position are given by $k_e = 3000\,(1 + \sin(0.2\pi t))$ N/m and $x_e = 0.02\,\sin(0.2\pi t)$ m.

These scenarios emulate practical conditions in tasks such as robotic ultrasound scanning or palpation, where soft tissue stiffness and contact location may change due to physiological motion or surface irregularities. Scenario 1 evaluates robustness to stiffness variation, Scenario 2 tests adaptability to moving contact points, and Scenario 3 represents the coupled, most challenging condition requiring comprehensive adaptability.

In this section, the simulation framework, including the robot and environment models, as well as three control methods with their respective parameter settings, has been introduced. Furthermore, three different scenarios are designed to evaluate the force tracking performance of the controllers under complex environments. These conditions enable a fair and comprehensive comparison of the proposed DOSMAC method against the traditional AC and the adaptive AVAC controllers.

4.2. Performance metrics

Several performance metrics are employed to quantitatively evaluate and compare the force tracking performance of the three control methods under complex environments.

To quantitatively evaluate the transient and steady-state force tracking performance of different controllers, three commonly used performance indices were employed: Overshoot (OS) Peak Force Error (PFE), and Root Mean Square Error (RMSE). These metrics are computed based on the measured contact force trajectory $F(t)$, the desired force F_d, and the key time instants of contact (t_c) and stabilization (t_s).

1) Contact time (t_c) and stabilization time (t_s):
   The contact time is defined as the instant when the measured force first exceeds a pre-defined threshold: $t_c = min\{t|F(t) \geq 0.5N\}$ The stabilization time t_s is determined as the earliest time after when the force enters and remains within a tolerance band $\pm 0.05F_d$ for a continuous dwell time $T_{dwell} = 0.4\,\mathrm{s}$.
2) Overshoot (OS):
   Overshoot quantifies the largest positive deviation from the desired force during the transient phase $[t_c : t_s]$:

$$OS = \max_{t\in[t_c:t_s]}\left(\frac{F(t-F_d)}{F_d} \times 100\%\right), OS \geq 0 \tag{40}$$

3) Peak force error (PFE):
   PFE measures the maximum absolute force deviation in the steady-state phase $[t_s : t_{end}]$:

$$PFE = \max_{t\in[t_s:t_{end}]}(F(t) - F_d) \tag{41}$$

4) Mean root square error (RMSE):

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}[F(t_i) - F_d]^2} \tag{42}$$

For a comprehensive evaluation, the global RMSE over the entire simulation is reported, which reflects the overall tracking performance.

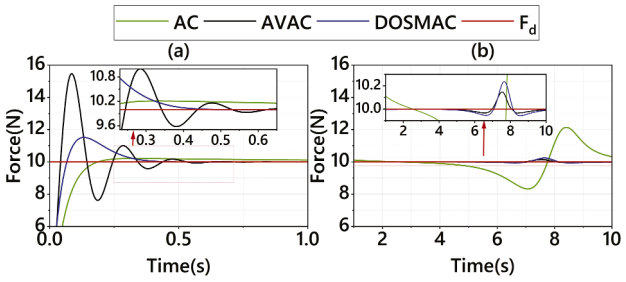


Fig. 3. Scenario 1 – Sinusoidal stiffness: contact force response. (a) transient response within 0–1 s; (b) steady state within 1–10 s.

Table 1
Performance metrics in simulation.

Scenario	Method	OS (%)	PFE (N)	RMSE (N)
Scen. 1	AC	2.1	2.14	0.99
	AVAC	54.7	0.49	0.60
	DOSMAC	15.1	0.49	0.42
Scen. 2	AC	24.3	2.51	1.81
	AVAC	59.4	0.50	0.65
	DOSMAC	19.2	0.49	0.44
Scen. 3	AC	27.4	3.54	2.36
	AVAC	59.6	0.47	0.61
	DOSMAC	18.9	0.49	0.43

Simulation results:

In this subsection, the simulation results of the three compared methods under different scenarios are presented. Furthermore, quantitative evaluation metrics are computed to compare the force-tracking performance of the three approaches in complex environments.

Fig. 3 illustrates the contact force tracking curves of the three control methods under scenario 1, where the red line denotes the desired force F_d. Fig. 3(a) shows the transient response within 0–1 s, while Fig. 3(b) presents the steady-state behavior over 1–10 s. For performance comparison, the evaluation metric of all three scenarios are summarized in Table 1.

As shown in Fig. 3, the three control methods exhibit distinct differences in force tracking under varying stiffness. In the transient phase (Fig. 3(a)), the AC method exhibits almost no significant overshoot during initial contact, but converges relatively slowly. In contrast, the AVAC method generates a large overshoot at the moment of contact, with the peak force considerably exceeding the desired value. The proposed DOSMAC method also presents a certain level of overshoot, but with smaller oscillations and faster convergence toward the steady state. During the subsequent steady-state phase (Fig. 3(b)), the AC method exhibits periodic fluctuations in force tracking, which are strongly influenced by stiffness variations. Both AVAC and DOSMAC can maintain the force close to the desired value; however, AVAC still exhibits noticeable oscillations, while DOSMAC achieves a more stable response.

The quantitative results in Table 1 further confirm these observations. The AC method has the smallest overshoot (2.1%) but suffers from a large peak force error (2.14 N) and a high RMSE (0.99 N), indicating poor steady-state performance. AVAC significantly improves steady-state accuracy with a PFE of 0.49 N and an RMSE of 0.60 N. However, its aggressive parameter adjustment results in a very large overshoot (54.7%), which compromises contact safety during the initial phase. By comparison, DOSMAC achieves a balanced performance with a moderate overshoot (15.1%) and superior steady-state accuracy (PFE = 0.49 N, RMSE = 0.42 N), outperforming the other two methods overall.

The differences can be attributed to the inherent mechanisms of each controller. For AC, although previous work (Jung et al., 2004) has demonstrated that setting the stiffness parameter $k = 0$ enables adaptation to environments with different stiffness, its fixed admittance

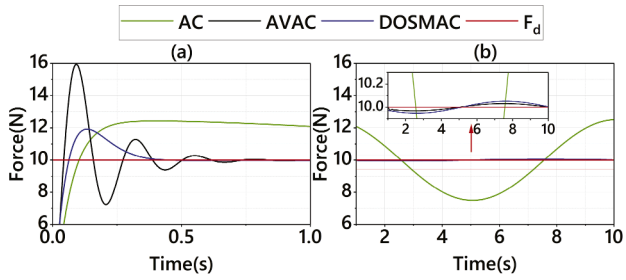


Fig. 4. Scenario 2 – Sinusoidal position: contact force response. (a) transient response within 0–1 s; (b) steady state within 1–10 s.

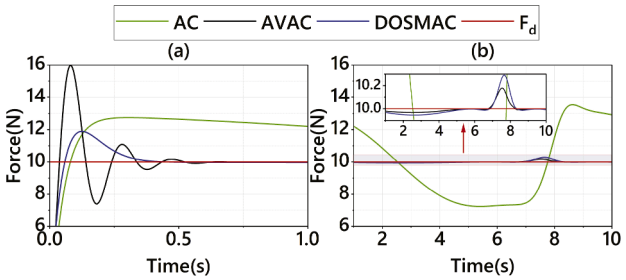


Fig. 5. Scenario 3 – Simultaneous Stiffness and Position Variation : contact force response. (a) Transient response within 0–1 s; (b) Steady state within 1–10 s.

parameters lack adaptability to time-varying stiffness, leading to poor steady-state accuracy. AVAC improves accuracy through adaptive damping adjustment, but its parameter update mechanism, which resembles integral action on the force error, causes strong overshoot at the initial stage. In contrast, DOSMAC leverages sliding mode control and disturbance compensation to effectively suppress uncertainties caused by varying stiffness while avoiding excessive parameter adjustment, thereby achieving a better trade-off between transient and steady-state performance.

Fig. 4 illustrates the contact force tracking curves of the three control methods under the condition of a time-varying environment position. The transient responses are shown in Fig. 4(a). Unlike Scenario 1, AC exhibits poorer adaptability when facing a time-varying environment position, with a larger overshoot in the initial stage compared to Scenario 1. Both AVAC and DOSMAC show behaviors similar to Scenario 1 in the initial phase, though with slightly higher overshoot. In the steady-state stage shown in Fig. 4(b), AC performs worse than in the varying stiffness scenario, failing to maintain the desired force and exhibiting large fluctuations. In contrast, AVAC and DOSMAC remain stable and maintain the desired force.

As shown by the performance indices in Table 1, AC has an overshoot of 24.3%, which is 22.2% higher than in Scenario 1. Its PFE reaches 2.51 N, and RMSE increases to 1.81 N. AVAC shows the largest overshoot at 59.4%, which is 5% higher than in Scenario 1. However, its steady-state performance remains stable, with a PFE of 0.5 N, nearly identical to the varying stiffness case, and an RMSE of 0.65 N, slightly higher than in Scenario 1 due to the increased overshoot. The proposed DOSMAC method achieves the smallest overshoot at 19.2%, the lowest among the three methods. Its RMSE is 0.44 N, slightly higher than in Scenario 1 but still the best among the three, demonstrating that DOSMAC provides the most favorable performance in the presence of environment position variations.

The results of Scenarios 1 and 2 indicate that under the designed simulation conditions, variations in environment position led to more pronounced performance degradation for AC compared to variations in environment stiffness. This suggests that the fixed admittance parameters of AC may make it more sensitive to position changes, whereas AVAC and DOSMAC maintained relatively stable performance.

To evaluate the force tracking performance of the three methods under more practical and complex conditions, Scenario 3 is designed, in which both environment stiffness and environment position vary simultaneously.

Fig. 5 presents the force tracking results of the three methods in Scenario 3. As shown, the results are generally similar to those of Scenario 2. In the transient phase (Fig. 5(a)), the AC method exhibits a large overshoot followed by a slow decrease in the contact force. The AVAC method demonstrates the largest overshoot, accompanied by oscillations for about 0.5 s before stabilizing around the desired force. In contrast, the proposed DOSMAC method shows the smallest overshoot and rapidly converges to the target force after the initial transient. Fig. 5(b) shows the steady-state response from 1 to 10 s. The AC method exhibits significant fluctuations, with the contact force varying between 7 and 14 N. Both AVAC and the proposed DOSMAC maintain stability around the desired force.

Table 2
Parameters of the force/torque sensor.

Parameter	Force (x–y–z)	Torque (x–y–z)
Range	±50.0 N	±10.0 N·m
Precision	±3.5 N	±0.2 N·m
Accuracy	±4.0 N	±0.3 N·m

The quantitative metrics of Scenario 3 are summarized in Table 1. AC shows the poorest overall performance, with an RMSE of 2.36 N and a PFE of 3.54 N, both worse than those of the two advanced methods. Its initial overshoot is 27.4%, which ranks in the middle among the three controllers. AVAC exhibits the largest overshoot of 59.6%, comparable to Scenario 2; however, its steady-state performance is superior, with an RMSE of 0.61 N and a PFE of 0.47 N, which is significantly better than AC. The proposed DOSMAC achieves the best overall performance, with the smallest overshoot of 18.9% and the lowest RMSE, indicating its superior balance between transient and steady-state behavior.

In summary, the evaluations across Scenarios 1–3 demonstrate that among the three methods, AC consistently delivers the weakest performance, primarily due to its fixed admittance parameters, which limit adaptability to environmental variations. AVAC benefits from its adaptive parameter update mechanism, leading to improved steady-state force tracking in time-varying environments; however, its reliance on accumulated force error for parameter tuning results in substantial overshoot during the initial phase, which is undesirable in applications such as robotic ultrasound scanning, where excessive contact force may risk patient safety. By contrast, the proposed DOSMAC leverages disturbance observer and sliding mode control enhancements, achieving the best overall performance when facing simultaneous stiffness and position variations.

5. Experiment validation

To verify the performance of the proposed method in a real robot system, a series of contact force tracking experiments was conducted involving varying environmental stiffness or position.

5.1. Experiment setup

The experiment platform is illustrated in Fig. 6. All experiments in this study were performed using a UR5e collaborative robotic manipulator (Universal Robots, Denmark), which features 6 degrees of freedom and a rated payload capacity of 5 kg. The UR5e is equipped with an integrated six-axis force/torque (F/T) sensor located at the robot's wrist (flange), enabling direct measurement of interaction forces and torques at the end-effector. The robot is rigidly mounted on a laboratory workbench under controlled ambient conditions. An ultrasound probe (DAX, Siemens Healthineers) is mounted on the end of the robot.

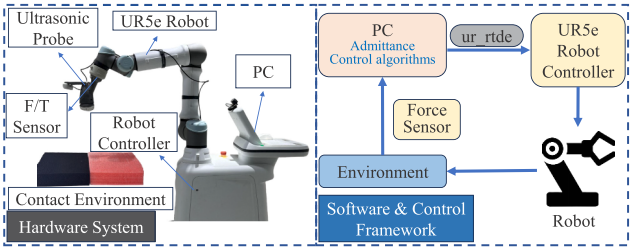


Fig. 6. Experiment platform and control framework for contact tasks.

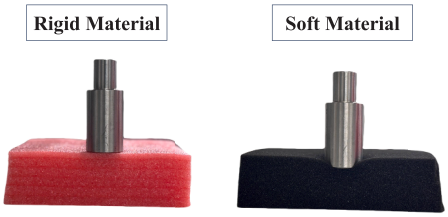


Fig. 7. Two types of testing material deformation under the same weight.

Robot control and data acquisition were executed on an external PC, which communicated with the robot controller via an Ethernet connection. The open-source ur_rtde Python interface was utilized to achieve real-time, low-level access to joint position, velocity, and force/torque data streams. The control loop and data acquisition were both executed with a cycle time of 2 ms (500 Hz), ensuring sufficient temporal resolution for dynamic force control tasks. The PC used in this experiment is equipped with an Intel(R) Core(TM) i5-10500T CPU running at 2.30 GHz.

The F/T sensor provides synchronous six-dimensional force and torque measurements during all experiments. The detailed specifications of the UR5e integrated force/torque sensor are listed in Table 2.

5.2. Experiment design

To evaluate the effectiveness and robustness of the proposed force control strategy, three experimental scenarios were designed under different interaction conditions, corresponding to the simulation studies.

Experiment I (Stiffness variation only): Two materials with different stiffness values were used to examine the controller's ability to maintain accurate force tracking under abrupt stiffness changes (Fig. 8). The two different materials are exhibited in Fig. 7; the stiffness of the two materials is around 900 N/m and 3500 N/m, respectively.

Experiment II (Constant Stiffness Curved Surface): A human abdominal phantom was used to evaluate the force tracking performance of the controller when interacting with a curved surface of approximately constant stiffness, approximately 500 N/m (Fig. 11).

Experiment III (Combined Stiffness and Position Variation): To assess the comprehensive performance of the controller in complex environments, materials with different stiffness values were attached to a curved surface (Fig. 13). This setting introduced both stiffness variation and position variation, thereby creating a more challenging interaction scenario.

5.3. Comparison methods and parameters

Three admittance control methods are conducted for comparison to address the advantages of the proposed DOSMAC. The parameters of each method is set as follows:

AC: Traditional admittance controller described in Eq. (5), and $m = 6, b = 900$ are used in the following experiments.

AVAC: Admittance controller with adaptive damping parameter described in Eqs. (38) and (39), the same admittance parameters m and b with AC are used, and is adopted for balance between large force oscillation and force tracking accuracy.

DOSMA: The proposed disturbance observer-based sliding mode admittance controller. The nominal environment stiffness $k_{e0} = 1000$ N/m is given. The rest of the parameters of the sliding mode controller are given as: $C = 5, k_s = 18, \eta = 5, \varphi = 0.1$. The DOB parameters $L_1 = 50$ and $L_2 = 750$ are given for fast and robust disturbance estimation.

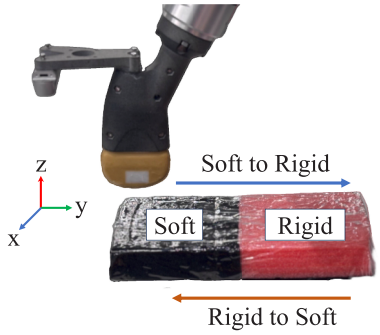


Fig. 8. Experiment I: Soft-rigid switching environment.

5.4. Experiment results

In this section, the experiment results with figures and tables are presented. To reduce the effect of chance, each experiment was repeated five times under identical conditions. The plotted curves correspond to the mean values across the five trials, and the discrete points represent the individual measurements, illustrating the variability across runs.

Experiment I: Soft-rigid switching environment

To realistically simulate the abrupt changes in tissue stiffness encountered during clinical ultrasound examinations, a soft-rigid switching environment experiment was performed. This experiment is intended to evaluate the adaptability and robustness of the proposed force control algorithms when the robot interacts with environments exhibiting sharp stiffness transitions, which are common in medical scenarios such as robotic palpation and ultrasound scanning.

At the start of each trial, the robot end-effector was positioned just in contact with the contact surface. Then, the robot moves with a constant velocity of 15 mm/s in y-axis according to a predefined trajectory. As illustrated in Fig. 8, there are two predefined trajectories (Soft-to-Rigid and Rigid-to-Soft) in this experiment. The movement of the robot in z-axis is totally driven by force control methods. This experiment provides a direct and practical assessment of the controller's ability to maintain stable contact force and adapt to sudden environmental changes, reflecting key challenges in real-world robotic medical interventions.

Figs. 9 and 10 present the force control results of Experiment I under the Rigid-to-Soft and Soft-to-Rigid scenarios, respectively. Table 3 presents the performance metrics of the three comparison methods in Experiment I.

For the *Rigid-to-Soft* case (Fig. 9), the AC method exhibits good transient performance in the initial stage, quickly converging to with only a small overshoot. However, during the stiffness transition, the maximum contact force deviation of AC reaches 2.36 N, after which it slowly returns to the vicinity of F_d. This observation confirms the theoretical derivation that setting in AC allows it to adapt to different fixed environment stiffness values; however, during transitions between environments of different stiffness, noticeable force deviations occur. The parameter adaptation rule of AVAC is based on the accumulation of force error. Consequently, in the initial stage, the accumulated force error during the force rise from 0 to leads to a larger overshoot. As shown in Table 2, AVAC's overshoot reaches 62.0% in the *Rigid-to-Soft* case, the largest among the three methods. Nonetheless, AVAC can stably converge to after the overshoot and shows a significant improvement over AC during stiffness transitions, with a post-initial PFE of 0.69 N, smaller

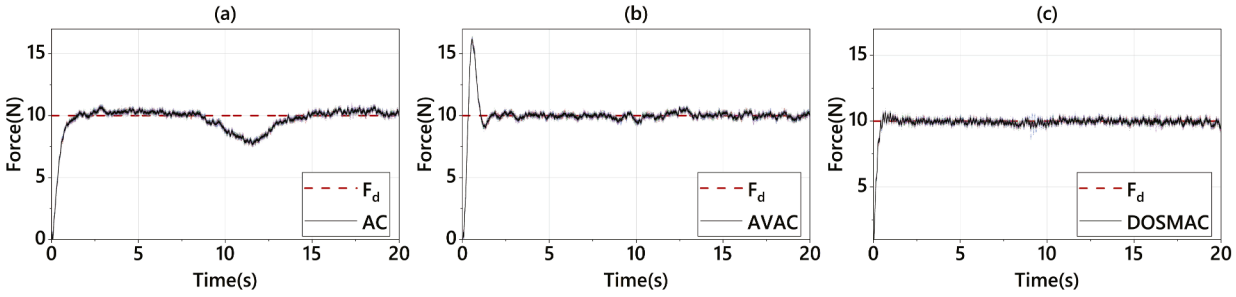


Fig. 9. Force control results of Experiment I (Rigid-to-Soft). Mean force trajectory (solid) and five individual trial measurements (markers).

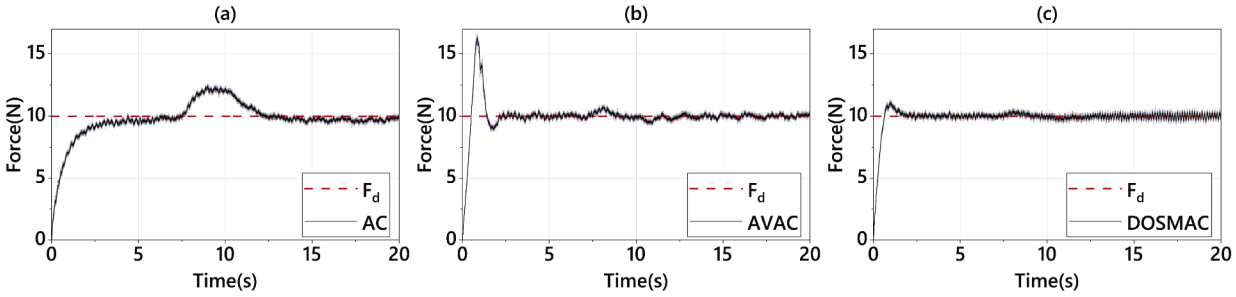


Fig. 10. Force control results of Experiment I (Soft-to-Rigid). Mean force trajectory (solid) and five individual trial measurements (markers).

Table 3
Comparison of three control methods in Experiment I.

Exp.	Method	OS (%)	PFE (N)	RMSE (N)
Exp I (R2S)	AC	8.3	2.36	1.51
	AVAC	62.0	0.69	1.35
	DOSMAC	6.4	0.64	0.86
Exp I (S2R)	AC	0.6	2.41	1.74
	AVAC	63.1	0.79	1.43
	DOSMAC	11.1	0.49	1.05

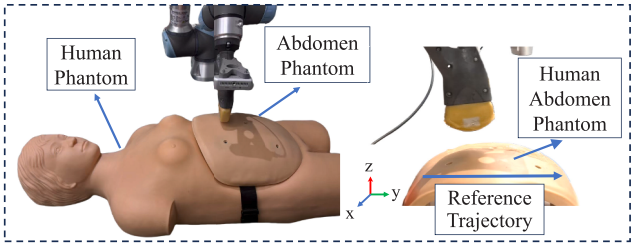


Fig. 11. Experiment II: Force control task on a human abdomen phantom.

than that of AC. For DOSMAC, the overshoot in the initial contact stage is 6.4%. In the subsequent steady contact stage, DOSMAC achieves a PFE of 0.64 N, the lowest among all three methods. Table 2 also reports the RMSE: AC, AVAC, and DOSMAC achieve RMSE values of 1.51 N, 1.35 N, and 0.86 N, respectively.

For the *Soft-to-Rigid* case (Fig. 10), due to the fixed admittance parameters, AC exhibits a longer rising time in the soft environment compared to the rigid one, but achieves the smallest overshoot (0.6%) in the transient period. During the subsequent stiffness transition, AC shows a trend where the contact force first increases and then returns to , with a PFE of 2.41 N in the process. AVAC's results indicate that in the initial stage, the reduced material stiffness leads to a longer rise phase and greater accumulated force error, resulting in a larger overshoot of 63.1%. The steady contact stage is similar to the *Rigid-to-Soft* case, with a smaller force error than the AC case. The proposed DOSMAC also presents a slightly larger overshoot (11.1%) than in the *Rigid-to-Soft* case. Nevertheless, DOSMAC achieves the smallest PFE and RMSE among the three methods during steady contact.

In summary, in *Experiment I*, where only the environment stiffness changes, the fixed-parameter AC exhibits large force errors during stiffness transitions. However, it can eventually return to the desired contact force. The parameter adaptation mechanism of AVAC enables better force error performance during stiffness changes; nevertheless, its error-accumulation-based adaptation results in a significant initial overshoot. The proposed DOSMAC, which integrates a disturbance observer and sliding mode control, effectively suppresses the force error induced by stiffness variations and achieves the smallest PFE and RMSE among the three methods, demonstrating its effectiveness in varying stiffness scenarios.

Table 4
Comparison of three control methods in Experiment II.

Exp.	Method	OS (%)	PFE (N)	RMSE (N)
Exp II	AC	54.4	3.91	2.83
	AVAC	124.3	0.57	1.98
	DOSMAC	25.1	0.61	1.17

Experiment II: Constant stiffness curved surface

The second experiment was conducted using a human abdomen phantom as the test object, as illustrated in Fig. 11. During the experiment, the robot was programmed to move the probe at a constant speed of 10 mm/s along a straight-line trajectory parallel to the y-axis, starting from one edge of the phantom and ending at the opposite edge. Throughout this motion, force control was activated exclusively on the z-axis, enabling the system to regulate the contact force between the probe and the phantom while accommodating anatomical surface undulations and material heterogeneity. This experiment is designed to simulate the typical procedure of robotic-assisted ultrasound scanning, where maintaining a stable contact force and adapting to complex tissue properties are crucial for achieving high-quality imaging and ensuring patient safety.

Fig. 12 and Table 4 present the force control results and performance metrics of the three control methods in Experiment II.

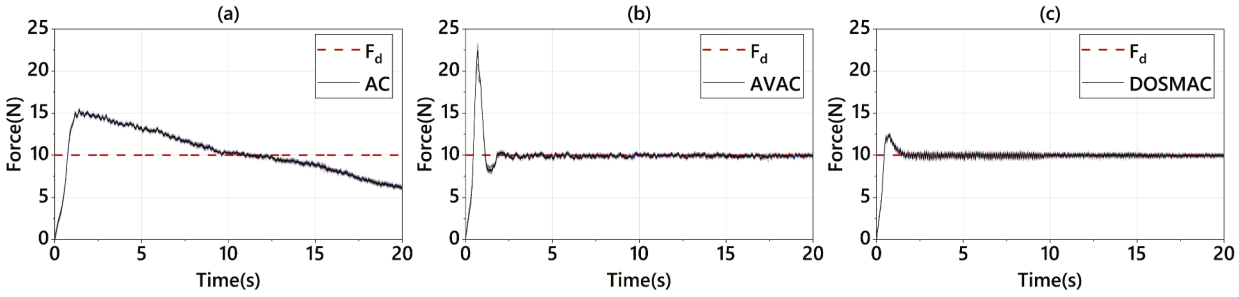


Fig. 12. Force control results of Experiment II (human abdomen phantom). Mean force trajectory (solid) and five individual trial measurements (markers).

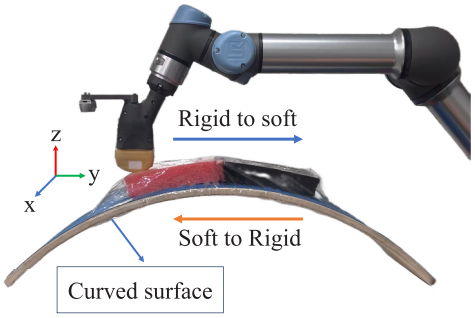


Fig. 13. Experiment III: Combined stiffness and position variation.

The results of Experiment II demonstrate that variation in environment position can cause the traditional fixed-parameter AC to lose stable force-tracking capability. Due to the upward movement of the environment position, AC exhibits a significant overshoot in the initial stage, followed by a slow tendency to recover toward F_d. After passing the middle section, the environment position begins to descend, and the contact force of AC starts to decrease. AVAC again shows a significant initial overshoot, reaching 124.3%, but, benefiting from its parameter adaptation mechanism, subsequently stabilizes near F_d. The corresponding PFE and RMSE are 0.57 N and 1.98 N, respectively. DOSMAC achieves the smallest overshoot of 25.1%, with a PFE of 0.61 N and an RMSE of 1.17 N, which is also the lowest among the three.

Experiment III: Complex environment with varying stiffness and position

To further evaluate the adaptability of the proposed force control algorithms in complex and realistic environments, experiment III was designed using a composite testbed. As shown in Fig. 13, two foam blocks with distinct stiffness values (black: softer, red: stiffer) were placed adjacent to each other on top of a curved base plate shaped as a sinusoidal surface. The entire setup was secured using a layer of plastic film to prevent displacement during the experiment.

The robot end-effector was commanded to follow a straight-line trajectory along the y-axis at a constant velocity of 15 mm/s. Only the z-axis motion of the end-effector was regulated by the force control algorithm, allowing the system to automatically adapt to both the varying surface height and the spatially varying contact stiffness. The primary objective is to assess the controller's ability to maintain consistent force tracking and adapt to simultaneous variations in both contact geometry and local stiffness.

Figs. 14 and 15 present the force control results of Experiment III under the Rigid-to-Soft and Soft-to-Rigid scenarios with environment position, respectively. Table 5 presents the performance metrics of the three comparison methods in Experiment II. Different from Experiment I and Experiment II, this experiment combines variations in the environment position and environment stiffness. For both *Rigid-to-Soft* and *Soft-to-Rigid* cases, AC exhibits large overshoots (35.5% and 43.4%, respectively) and shows contact force variations during stiffness transitions similar to those in Experiment I—an increase in contact force for *Soft-to-Rigid* and a decrease for *Rigid-to-Soft*. However, after the stiffness transition is completed, due to the continuous downward movement of the environment, AC can no longer maintain the contact force near , resulting in a sustained decrease in force. The PFE after the initial contact reaches 4.62 N and 4.90 N, respectively, with the maximum deviation occurring at the very end of the contact. The corresponding RMSE values are 2.44 N and 2.85 N.

Table 5
Comparison of three control methods in Experiment III.

Exp.	Method	OS (%)	PFE (N)	RMSE (N)
Exp III (R2S)	AC	35.5	4.62	2.44
	AVAC	61.8	0.90	1.25
	DOSMAC	6.4	0.63	0.86
Exp III (S2R)	AC	43.4	4.90	2.85
	AVAC	64.7	1.00	1.46
	DOSMAC	14.4	0.54	1.06

For AVAC, similar to Experiment I, a significant overshoot is observed in the initial contact stage, followed by convergence to a force near the desired value. Moreover, when facing simultaneous changes in environment stiffness and position, AVAC shows a significant improvement over AC, with PFE values of 0.90 N and 1.00 N, and RMSE values of 1.25 N and 1.46 N for *the Rigid-to-Soft and Soft-to-Rigid transitions*, respectively.

Benefiting from the disturbance observer's ability to capture variations in both environment stiffness and position, combined with sliding mode control, the proposed DOSMAC method achieves the best performance in both R2S and S2R scenarios, with the smallest overshoot (6.4% and 14.4%), PFE (0.63 N and 0.54 N), and RMSE (0.86 N and 1.06 N) among the three methods, as summarized in Table 5.

The results of Experiment III demonstrate that variation in environment position can cause the traditional fixed-parameter AC to lose stable force-tracking capability. Except for the larger overshoot observed in the initial contact phase, AVAC's parameter adaptation function enables improved adaptability to changes in both environment stiffness and position after the initial contact is established. The proposed DOSMAC, through the combined use of a disturbance observer and sliding mode control, is able to compensate for and suppress disturbances induced by variations in environment stiffness and position, thereby achieving the best force-tracking performance among the three methods.

The results demonstrate consistent behavior across repeated trials, with limited variability in force-tracking accuracy, confirming the repeatability of the proposed control scheme under the tested phantom conditions.

6. Discussion

The results of this study consistently demonstrate that the proposed disturbance-observer based sliding mode admittance control (DOSMAC)

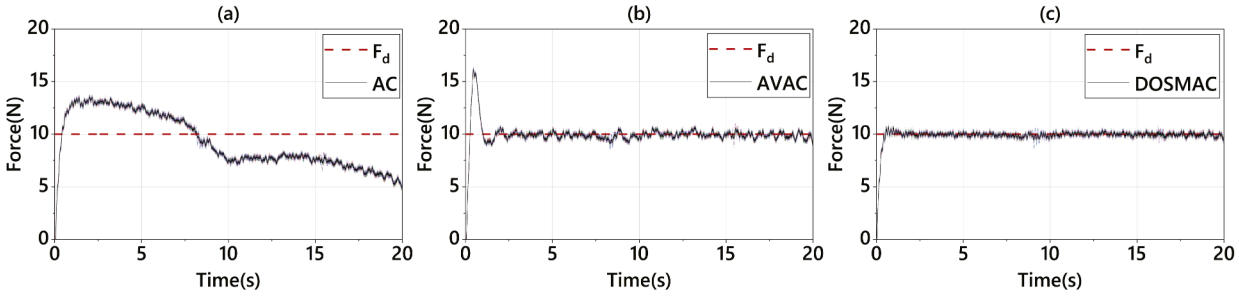


Fig. 14. Force control results of Experiment III (Rigid-to-Soft). Mean force trajectory (solid) and five individual trial measurements (markers).

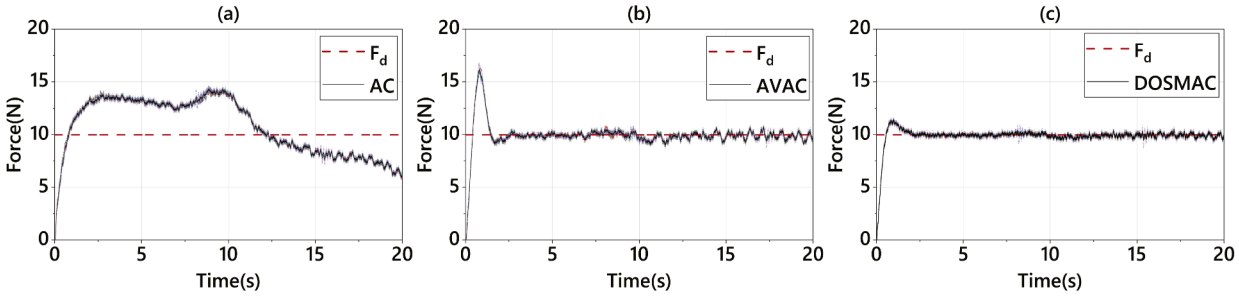


Fig. 15. Force control results of Experiment III (Soft-to-Rigid). Mean force trajectory (solid) and five individual trial measurements (markers).

achieves superior force-tracking performance compared with conventional admittance control (AC) and adaptive variable admittance control (AVAC). From a theoretical perspective, AC with zero virtual stiffness guarantees stability only under a constant environment stiffness; however, its inability to compensate for time-varying stiffness and position errors results in significant tracking degradation, as confirmed by both simulations and experiments. AVAC improves steady-state accuracy by adjusting damping parameters based on force error, yet its adaptation mechanism effectively introduces an integral action that amplifies error accumulation during initial contact, leading to large overshoot. In contrast, DOSMAC reformulates the dynamics by lumping all unknown effects of varying stiffness and position estimation errors into a disturbance term, which is then estimated and compensated through a disturbance observer. Together with the sliding mode control law, this ensures that the sliding variable converges to zero even in the presence of uncertainties.

For the proposed DOSMAC, the sliding mode framework enables fast convergence of the force error, while the disturbance observer reduces the impact of unmodeled dynamics, thereby reducing reliance on large switching gains and mitigating chattering. Moreover, steady-state accuracy is improved by compensating for the time-varying stiffness and position-related terms that degrade the performance of AC and AVAC, which explains the lower RMSE observed across complex interaction scenarios. These results indicate that the theoretical stability guarantees established through Lyapunov analysis are effectively realized in practice.

It is worth noting that the experimental results show that both AVAC and DOSMAC exhibit larger overshoot in the initial contact phase when interacting with softer environments. This phenomenon can be explained by examining the underlying control mechanisms. For AC, which essentially relies on the stiffness-dependent force–displacement relationship, harder environments produce sharper force changes for the same displacement, making the system more prone to transient overshoot; in softer environments, the same displacement induces only small force variations, resulting in smaller overshoot. In contrast, AVAC and DOSMAC are active compensation controllers. In soft environments, the slower rise of the contact force is interpreted as a large tracking error, which activates stronger parameter adaptation in AVAC or a faster sliding-mode convergence in DOSMAC, in both cases driving the system to respond more aggressively. Consequently, overshoot becomes larger in softer environments. This highlights a fundamental difference between passive and active control strategies: passive schemes, such as AC, tend to overshoot more in hard environments, whereas active schemes, such as AVAC and DOSMAC, are more prone to overshoot in soft environments.

In the context of robotic ultrasound, these findings have important implications. Excessive overshoot, as observed with AVAC, results in transiently large probe–tissue contact forces that may cause discomfort or harm, while steady-state errors impact probe stability and image quality. The ability of DOSMAC to combine small overshoot with high steady-state accuracy, therefore, represents a critical advantage for ensuring both patient safety and reliable imaging outcomes.

The proposed approach is also consistent with broader trends in robust force control, in which disturbance-rejection structures and nonsmooth regulation are increasingly used to handle uncertainties in robot–environment interaction. The integration of observer-based disturbance estimation with a sliding-mode mechanism also resonates with model-free or minimally modeled adaptive control strategies designed for time-varying interaction conditions.

Several practical considerations should be noted. First, although the probe orientation is fixed in this study, improper orientation may affect imaging quality. In clinical scenarios, real-time surface-normal following will be essential, and integrating DOSMAC with orientation control will be addressed in future extensions. Second, potential time delays arising from sensing and low-level servo loops are not explicitly modeled, following the standard assumption that such delays are small relative to the controller bandwidth. Delay-aware analysis or compensation could be incorporated in future extensions, drawing on recent advances in time-varying delay systems. Third, phantom-based validation provides repeatable conditions but does not fully capture the variability of in vivo tissues; further evaluation in human–robot interaction scenarios will be needed to assess clinical applicability. Finally, a single set of controller parameters was used across all experiments, demonstrating robustness to parameter variation and suggesting suitability for clinical deployment, where patient and tissue dependent retuning is undesirable (Algorithm 1).

Consistent force regulation is essential for maintaining reliable acoustic coupling and image quality in clinical ultrasound procedures, and the robustness of the proposed framework suggests potential benefits for safe and repeatable scanning. The controller can be integrated with ultrasound imaging pipelines to enable image-guided or quality-aware motion adaptation in future systems. Although validation was performed on phantom models, further studies in more realistic human–robot interaction scenarios will be explored to assess clinical applicability

Overall, the combination of small overshoot, low steady state error, and strong robustness to stiffness and surface-motion variations highlights the potential of the proposed DOSMAC framework for safe, repeatable, and high-quality robotic ultrasound force regulation.

7. Conclusion

This work has addressed the challenge of robust force-tracking control in environments with unknown, time-varying stiffness and position, conditions under which conventional admittance-based methods often experience significant performance degradation. To overcome these limitations, a disturbance-observer based sliding mode admittance control (DOSMAC) framework was developed, where uncertainties caused by variations in environmental stiffness and estimation errors of the equilibrium position were reformulated as a lumped disturbance. Assuming this disturbance varies slowly with a negligible derivative, it can be effectively estimated and compensated through a disturbance observer, while the sliding mode component guarantees convergence of the force error.

The proposed approach was validated through both simulation and physical experiments conducted on phantom models with environmental stiffness in the range of approximately 500–3500 N/m. Across all tested scenarios, DOSMAC consistently achieved more accurate force tracking than conventional admittance control (AC) and adaptive variable admittance control (AVAC), combining low steady-state error with limited overshoot, even under complex interaction conditions. These findings confirm that the stability properties predicted by the Lyapunov analysis are realized in practice and demonstrate the effectiveness of the slowly varying disturbance assumption in modeling uncertainties during robotic ultrasound interactions.

While the present validation was restricted to phantom experiments with empirically tuned parameters, and the slow-disturbance assumption may be challenged in more dynamic environments, the results indicate that DOSMAC provides a reliable and practical framework for robust force control. By enabling accurate tracking across a wide range of stiffness, the method offers a promising foundation for advancing robotic ultrasound toward safe and consistent operation in clinical settings.

CRediT authorship contribution statement

Sen Li: Writing – original draft, Validation, Software, Methodology, Investigation, Data curation; **Qi Chen:** Validation, Software, Resources; **Weihua Li:** Funding acquisition, Formal analysis; **Mingfeng Wang:** Validation, Conceptualization; **Sungsung Pang:** Resources, Project administration; **Kai Wu:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.conengprac.2026.106764

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